Status	Finished
Started	Sunday, 25 October 2026, 11:14 PM
Completed	Sunday, 25 October 2026, 11:23 PM
Duration	9 mins 6 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search stops
- c. The list is sorted
- d. The search continues from the beginning

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 89 and 94
- b. 94 and 99
- c. 90 and 100
- d. 90 and 99 ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search requires the list to be sorted.
- b. Linear search can be applied to both sorted and unsorted lists. ✓
- c. Linear search is more efficient than binary search on large lists.
- d. Linear search divides the list into halves.

Question **4**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The search stops
b. The middle element is chosen randomly
c. The lower middle element is chosen as the middle element ✗
d. The higher middle element is chosen as the middle element

Question **5**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The list is sorted
b. The search ends successfully ✓
c. The search continues in the right sublist
d. The search continues in the left sublist

Question **6**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓
b. log n
c. n/2
d. 1

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Dynamic search
b. Linear search ✓
c. Hashing
d. Binary search

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Quick search
b. Linear search ✓
c. Bubble search
d. Merge search

Question **9**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Binary ✓
b. Hash
c. Linear
d. Both (1) & (3)

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Must use a sorted array
c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O(log n)
c. O(1)
d. O👎 ✓

Question **12**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(n log n)
b. O(log n)
c. O(1) ✓
d. O👎

Question **13**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the element is found in the list ✓
b. When the list contains duplicate elements
c. When the search algorithm finishes
d. When the list is sorted

Question **14**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 100
c. 1000
d. 10 ✓

Question **15**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Insertion

b. Selection

c. Bubble ✓

d. Complexity